

Chapter 5 — Principles of CT and Radiographic Imaging

- **1895 — Wilhelm Röntgen** discovered x-rays incidentally while experimenting with cathode rays; the first radiograph, of his wife's hand with a ring (Fig. 5.1), was **published in 1896**, and diagnostic radiology began at once.
- Radiographs moved from **hard-copy film stored in file rooms to digital acquisition**: images are now processed to improve quality and reduce noise, stored electronically and viewed by several physicians at once.
- Fluoroscopy moved from **continuous exposure with poor images to low-dose pulsed technology with last image hold**, giving better images at lower dose.
- Computer image processing produced **computed tomography (CT)**; artificial intelligence (AI) is the next expected advance.

RADIATION BIOLOGY AND SAFETY

- The hazard was not recognised at first. **Clarence Dally**, an employee of Thomas Edison, repeatedly placed his hand in the beam while developing the fluoroscope → erythema, ulceration, hair loss → carcinoma → amputation of the left arm and four fingers of the right hand → death from metastatic cancer in 1904, aged 39: **the first known death from radiation exposure**. Edison afterwards declared himself afraid of x-rays.
- Governing principle: **ALARA — as low as reasonably achievable**.

Mechanism of injury.

- Absorbed ionising radiation damages cells by **direct DNA ionisation** or **indirect damage from free radicals**; depending on degree and site, the cell undergoes apoptosis or attempts DNA repair.

DETERMINISTIC VERSUS STOCHASTIC EFFECTS (FIG. 5.3)

	Deterministic	Stochastic
Threshold	Yes — no effect below it	None — linear no-threshold model
What rises with dose	Severity	Probability of disease, cumulatively
Mechanism	Sufficient cellular damage (cell loss)	Mutation in a proto-oncogene or tumour suppressor gene
Examples	Erythema, depilation, sterility, anaemia, cataract	Cancer
Safe dose	Below the threshold	No safe dose ; every exposure adds risk

Evidence of harm.

- **Hiroshima survivors** (high dose): radiation burns and increased leukaemia and solid cancers — bladder, breast, lung, brain, ovary and thyroid, among others.
- Diagnostic doses are far lower, yet **medical diagnostic radiation is estimated to cause 1.5–2% of cancers in the United States**.
- Health care workers: radiologists have **more leukaemia** than other specialists; physicians using fluoroscopy have **more brain tumours**. Other effects: cardiovascular disease, thyroid disease, inflammatory conditions and **cataract** — interventional cardiologists have a markedly raised risk of posterior lens opacity.

Dose units.

RADIATION DOSE QUANTITIES

Quantity	Unit	Definition	Use
Absorbed dose	Gray (Gy)	Ionising energy absorbed by tissue	Physical dose
Equivalent dose	Sievert (Sv)	Absorbed dose × radiation weighting factor; factor = 1 for medical x-rays , so equivalent = absorbed	Adjusts for radiation type
Effective dose	Sievert (Sv)	Equivalent dose × tissue weighting factors (tissue susceptibility)	Annual exposure limits and occupational monitoring

- Imaging use keeps rising despite low-dose developments, so **a risk-benefit analysis precedes every ionising study**. ALARA in practice: low-dose protocols, alternative studies, avoiding ionising radiation where possible — reducing the patient's cumulative dose and often the surgeon's and staff's.

KEY POINTS: RADIATION BIOLOGY AND SAFETY

- Radiography, fluoroscopy and CT use high-energy photons, which cause deterministic and stochastic effects.
- Under the linear no-threshold model, stochastic risk is proportional to dose and no amount is risk-free.
- Follow ALARA: use only the radiation necessary for diagnosis and management.

Concept — *Deterministic effects have a threshold and scale in severity; stochastic effects have no threshold and scale in probability. Because cancer risk follows the linear no-threshold model, there is no safe dose, and ALARA follows directly from that.*

RADIOGRAPHIC STUDIES

Radiographic Radiation Physics

- Diagnostic x-rays lie at 10^{16} – 10^{20} Hz in the electromagnetic spectrum — enough energy to ionise, hence "ionising radiation".
- High frequency lets x-rays pass through tissue; **differential absorption between tissues creates contrast**, which is the basis of the image.
- Generation: vacuum tube → potential difference across cathode and anode → electrons accelerate from **cathode to anode** → strike the anode plate, **usually tungsten** → x-ray emission.
- Film detection: x-rays strike an intensifying screen exposing the film → developed → **black where photons arrive, white where radio-opaque material attenuates**.
- Digital detection — **indirect**: x-rays stimulate a photoluminescent material whose fluorescence is read by a digital receiver; **direct**: a material excited directly by x-ray photons, no fluorescent step. Both yield an electrical charge read like a digital camera sensor.

Chest Radiography

- **Chest x-ray (CXR) is the commonest radiograph in hospital**: respiratory and cardiac disease; preoperative clearance (not always needed); high sensitivity for haemothorax, pneumothorax, atelectasis, effusion, infiltrates, infection and trauma; oncological staging and surveillance; **position of lines and tubes** (chest drain, endotracheal and gastric tubes, central venous lines).
- Dose: **0.07 mSv**.

- **Technique:** posteroanterior, upright, full inspiration; hands on hips to move the scapulae laterally; remove lines and tubes from the lung field where possible; lateral, oblique and expiratory views as adjuncts; portable film if immobile.
- **Limitations:** poor positioning, motion, rotation or altered exposure impair reading; subtle day-to-day change is easily missed; false positives prompt unnecessary intervention and false negatives miss disease — **CXR augments, never replaces, history and examination.**

Plain Abdominal Radiography

- The **kidney-ureter-bladder (KUB)** film (Fig. 5.4: kidneys, psoas shadows, liver tip, spleen, gastric gas): diagnosis and follow-up of **radio-opaque stones**; before shock-wave lithotripsy (SWL); pre- and post-contrast film; postoperative ileus, obstruction and free air; confirms line, tube, stent and drain position; foreign bodies; vertebral and pelvic fractures in trauma.
- Dose: **0.7 mSv per film; 1.4 mSv for a standard two-view abdomen.**
- **Technique:** supine anteroposterior (AP) film **from the upper renal margin to the inferior pubic rami**; obliques separate overlapping structures; **upright film is most sensitive for free intraperitoneal air.**
- **Limitations:** cheap and low dose but **largely supplanted by CT**, which gives the same information more accurately. KUB is two-dimensional, so overlying structures hide pathology: its sensitivity for stones, the acute abdomen and bowel obstruction is poor and variable, whereas CT is three-dimensional (3D) and detects **>95% of stones.**

Static Cystography

- Contrast instilled into the bladder shows **extravasation, filling defects** (stone, foreign body, haematoma, tumour), reflux, neurogenic bladder, diverticula, fistulas and leaks.
- **Trauma:** for suspected bladder injury the **American Urological Association (AUA) recommends plain or CT cystography**; CT cystography is used more often, but static cystography has **similar sensitivity.**
- Postoperative check of **genitourinary (GU) anastomotic integrity.**
- After radical prostatectomy: **bladder neck-to-pubic symphysis (BNPS) ratio** = distance from the superior border of the symphysis to the bladder neck ÷ total symphyseal height. **A lower ratio predicts better continence** (Fig. 5.5: continent patient 0.36; incontinent patient with a low-lying bladder neck 1.1).
- **Technique:** supine scout film → Foley catheter if not indwelling → **at least 300–350 mL dilute contrast by gravity** → clamp → AP film of full bladder → obliques (posterior diverticula, fistulas, leaks) → drain → **postdrainage film** (extravasation hidden behind dense contrast).
- **Limitations:** two-dimensional; sensitivity depends on **volume instilled** and operator skill; **active infection risks pyelonephritis or sepsis.**

KEY POINTS: RADIOGRAPHIC STUDIES

- X-rays are generated by accelerating electrons into an anode metal.
- KUB detects radio-opaque stones, bowel complications and stent and drain position.
- Plain films cost less and deliver less radiation than conventional CT.
- Static cystography needs baseline, filling and postdrainage films and assesses bladder, urethral and anastomotic integrity.

Concept — *Every plain study trades information for dose and cost against CT: projection loses depth and hides overlying pathology, so plain films survive where a narrow question (stone follow-up, device position, a leak) does not justify CT.*

FLUOROSCOPY

- Live x-ray imaging for procedures needing real-time feedback: **renal access for stones, nephrostomy and ureteric stent placement, InterStim (sacral neuromodulation) implantation, retrograde pyelography, cystography and voiding cystourethrography (VCUG).**
- Intraoperatively the **surgeon takes part in choosing machine settings** to keep dose to ALARA.

Fluoroscopy Physics

- X-ray production is as in radiography; detection differs. Early systems used a **fluorescent screen**; modern systems use an **image intensifier or a flat panel detector**.
- **Flat panel detector** — works like a conventional digital radiography detector.
- **Image intensifier** — x-rays strike the **input phosphor** (converts to a wavelength suited to the photocathode) → **photocathode** releases electrons → electrons **accelerated across a vacuum** by an applied potential → strike the **output phosphor** at the anode → converted back to light and recorded.
- Brightness gain has two sources: **electron acceleration** (energy gain) and **focusing electrons onto a smaller area**.

Effect of Machine Settings

PEAK KILOVOLTAGE VERSUS TUBE CURRENT

	Peak kilovoltage (kVp)	Tube current (milliamperes, mA)
Changes	Photon energy — maximum energy in kiloelectron volts (keV) proportional to kVp	Photon number
Dose	Rises exponentially (other factors constant)	Rises linearly
Penetration	Increased (higher frequency)	Unchanged
Image contrast	Decreased	Increased

Retrograde Pyelography

- Fluoroscopic study of the upper tract. A CT urogram gives much of the same information, but the retrograde pyelogram is **convenient intraoperatively**: maps the collecting system for retrograde intrarenal surgery and aids percutaneous access.
- **Chosen when intravenous (IV) contrast is hazardous**: contrast allergy or renal dysfunction.
- Evaluates haematuria, obstruction, trauma and tumour; detects **extravasation** (Fig. 5.6: calyceal leak at a partial nephrectomy site); shows **peristalsis**, which a CT urogram cannot; performed and read in real time by the surgeon.
- **Technique**: dorsal lithotomy → cystoscopy identifies the ureteric orifice → catheterise → **gentle injection of dilute contrast (often 50%)** — over-opacification hides small filling defects → fluoroscopic monitoring and mapping.
- **Air instead of contrast** reduces radiation but impairs later ultrasound and, very rarely, causes **air embolism** if injected under high pressure.
- **Limitations**: usually needs anaesthesia and an operating theatre (cost, complexity); more invasive than CT urography; risk of **iatrogenic ureteric injury**; **over-pressurisation causes backflow — pyelovenous, pyelotubular, pyelosinus, pyelolymphatic** (Fig. 5.7) → bacteraemia and sepsis.

Voiding Cystourethrography

- Static cystography plus **voiding-phase fluoroscopy**: sphincter and urethral function and **competence of the ureterovesical junction**.
- **Gold standard for vesicoureteral reflux (VUR)** and recurrent urinary tract infection (UTI). Also: posterior urethra (posterior urethral valves, diverticula), bladder (diverticula, tumours, leaks), ureters (ureterocele, diverticula), kidneys (when reflux present); neurogenic bladder, incontinence, bladder outlet obstruction, voiding difficulty.

GUIDELINES: IMAGING AFTER FEBRILE UTI IN CHILDREN

Body	Population	First test	VCUG
European Association of Urology (EAU)	Children with febrile UTI	Renal–bladder ultrasound	If ultrasound abnormal or age <12 months
American Academy of Pediatrics (AAP)	Age 2–24 months	Renal–bladder ultrasound	Withheld unless ultrasound suggests increased VUR risk

- **Technique**: supine scout → catheter by size — **infants 5 French (Fr) feeding tube; young children 6–8 Fr; adults 14–16 Fr** → fill by gravity, imaging early filling → fill to **age-appropriate capacity (Box 5.1)** → bilateral obliques (vesicoureteral junctions, prostatic urethra) → remove catheter → void under live fluoroscopy → postvoid film. **Repeat fill–void cycles** increase sensitivity in high-risk patients.

BOX 5.1 AGE-DEPENDENT BLADDER CAPACITY FORMULAS

Age (years)	Capacity (mL)
<2	Weight (kg) × 7
2–14	(Age in years × 30) + 30
>14	300–500

Data modified from ACR–SPR (American College of Radiology–Society for Pediatric Radiology) Practice Parameter, American College of Radiology, 2019.

- Purpose: the **target fill volume** for VCUG. Pattern: infants by **weight**, children by **age**, adolescents a fixed adult range. Trap: the age formula applies only from 2 to 14 years.
- **Limitations**: invasive → distress for patient and family; a substantial share of outpatient orders are never completed; **sterile urine required** (risk of seeding the upper tract); impossible without catheterisation (meatal or urethral stenosis, false passage, urethral disruption); **spinal cord injury above T6 → autonomic dysreflexia** on distension, may need anaesthesia; children are more radiosensitive and have longer life expectancy for cancer to develop.

Retrograde Urethrography

- Evaluates the **anterior urethra**: strictures, suspected trauma, postoperative setting; also diverticula, tumours, fistulas, foreign bodies. Alone or combined with cystography or VCUG; **performed before urethral stricture repair**; shows distal anatomy other modalities miss.
- **Technique**: **45° oblique, right shoulder down** → scout → flushed catheter inserted **about 2 cm into the fossa navicularis** → balloon inflated with **about 2 mL sterile water** (occludes; overinflation injures the urethra) → penis on traction to straighten the urethra → gentle contrast → anterior urethra filled to the external sphincter → further injection passes the sphincter into the prostatic urethra and bladder. Adult volume **about 20–30 mL**. With a Foley already in place, a **5 Fr feeding tube alongside it in the fossa** gives a pericatheter urethrogram.
- **The prostatic urethra does not normally distend on retrograde injection** — not to be read as stricture.

- **Limitations:** the posterior urethra is inadequately shown → add cystography or VCUG in the same session; catheter-related limitations as for VCUG.

RETROGRADE URETHROGRAM VERSUS VCUG

	Retrograde urethrogram	VCUG
Segment shown best	Anterior urethra	Posterior urethra, bladder, reflux
Direction of contrast	Retrograde from meatus	Antegrade during voiding
Classic use	Stricture mapping, trauma	VUR, posterior urethral valves

Loopography

- Retrograde contrast via the **stoma into the bowel segment** of a urinary diversion → assesses diversion and upper tracts.
- **Refluxing uretero-intestinal anastomosis** → contrast ascends the ureters → filling defects, obstruction. **Non-refluxing** → upper tracts not shown; an antegrade study is needed.
- Indications: haematuria, recurrent infection, pain, declining renal function; surveillance of upper tract neoplasia; anastomotic stricture; diversion patency.
- **Safe in mild-moderate contrast allergy or renal impairment** — minimal systemic absorption from bowel.
- **Technique:** supine scout → catheter through stoma into the loop → instil contrast → films during and after filling → **obliques** to show the whole upper tract → **postdrainage film** for obstruction.
- **Limitations:** balloon overinflation in a small-bowel segment can disrupt mucosa; overlapping loops; **excessive pressure** → **pyelonephritis**; non-refluxing anastomoses → no upper tract; **continent pouches** — **small tumours missed** (irregular walls, large contrast volume).

Fluoroscopic Renal Access

- Access to the collecting system for drainage in infection or obstruction (mostly by interventional radiology) and before stone surgery.
- For percutaneous nephrolithotomy (PCNL), ultrasound-guided access is increasing but is harder for **upper pole access (ribs), obesity and staghorn calculi**; **fluoroscopy remains the commonest technique**.

FLUOROSCOPIC RENAL ACCESS TECHNIQUES

Technique	Sequence	Distinguishing feature
Bull's-eye	Identify target calyx → image intensifier 15–30° towards surgeon → needle end-on as a dot over the calyx → advance under continuous fluoroscopy → rotate C-arm away from surgeon to judge depth → advance to loss of resistance (collecting system) or stone resistance	Needle kept parallel to the beam
Laser direct alignment radiation reduction technique (DARRT)	Bull's-eye positioning with pulsed fluoroscopy → laser on image intensifier or flat panel aligned with needle → insertion under laser guidance and intermittent pulsed images	With endoscopic combined intrarenal surgery, radiation cut substantially versus conventional access
Triangulation	C-arm at 0° to locate calyx → skin puncture medial to the posterior axillary line (prone) → C-arm 30° caudal for upper pole, 30° cranial for lower pole → needle aimed at the third corner (calyx) → alternate C-arm positions for depth and trajectory	Two views triangulate depth

- Robotic, electromagnetic-guidance and fusion access are in development but not yet common.

Specialty Fluoroscopy

- **Cavernosography** (corporal leak in erectile dysfunction or trauma), **vasography** (vas cannulated, contrast maps the system and checks patency), and **sacral nerve stimulator placement**.

Special Fluoroscopic Considerations

Three-Dimensional Fluoroscopy (Cone Beam Computerised Tomography)

- Surgeons normally rebuild 3D anatomy mentally from two-dimensional images (**cognitive fusion**). Cone beam CT: C-arm rotates (continuous or pulsed) → many images → **filtered back projection** reconstruction as in CT; needs a specialised table for circumferential imaging.
- Uses: planning and guiding needle access; **intraoperative detection of residual stones** → immediate improvement of stone-free rate.
- CT scanners with a fluoroscopy mode: poorer surgeon access, harder sterile field, **higher dose**.

KEY POINTS: FLUOROSCOPY STUDIES

- Fluoroscopy gives real-time anatomy during diagnostic and therapeutic procedures.
- Retrograde pyelography maps the upper tract intraoperatively and is usually safe with contrast allergy and renal failure.
- VCUG captures voiding-phase images and is the standard test for VUR.
- Retrograde urethrography evaluates the anterior urethra for stricture, leak, diverticulum and fistula.
- Loopography evaluates the urinary tract after diversion.
- Fluoroscopy establishes renal access for drainage or before stone treatment.

Concept — *Every retrograde contrast study carries the same two liabilities* — **catheterisation and pressurised contrast into a possibly infected system** — *so sterile urine, gentle low-pressure filling and choosing the direction (retrograde for the anterior urethra, voiding for the posterior) recur across all of them.*

Fluoroscopic Radiation Reduction Techniques

- Fluoroscopy use varies widely between institutions, machines and surgeons; dose reduction benefits patients and staff. The United States Food and Drug Administration issued a white paper urging reduction. Summary of measures (Fig. 5.8): lead gloves, curtain, vest, glasses, thyroid shield and cap; instruments out of the field; collimation; source below the patient; low-dose setting and lower pulse rate; distance from the source; last image hold; recording fluoroscopy time.

Fluoroscope Adjustments

- **Time. Dose is proportional to fluoroscopy time.** Avoid unnecessary images and studies; use **last image hold** with intermittent spot images; prefer **direct endoscopic vision**. Ureteroscopy and PCNL have been performed **with no radiation**, using visual and tactile cues, direct vision and ultrasound. Every activation must be justified.
- **Collimation.** Narrowing the aperture to the region of interest; **halving the aperture diameter reduces the exposed area fourfold**. Excluding radiodense areas lowers patient dose and improves contrast and image quality.
- **Low-dose mode.** Roughly **halves the current (mA)**, and **dose falls by a similar amount**; contrast falls but images remain adequate for most urology.
- **Pulsed fluoroscopy.** Continuous fluoroscopy = **30 pulses per second (pps)**. Theoretical reduction at 8, 4 and 1 pps = **73%, 87% and 97%**; the **real saving is smaller because of the ramp and tail effect** of each pulse (Fig. 5.9: modern pulsed units have shorter ramp and tail than conventional pulsed units).

- *Dose tracking and reporting.* Recording fluoroscopy time builds awareness, sets goals and measures reduction strategies; surgeon feedback alone reduces fluoroscopy time. **Surgeons should wear dosimeters.**

Shielding

- Lead aprons, usually **0.25–0.50 mm lead equivalent**, absorb most incident radiation; add a **thyroid shield**.
- Standard shielding leaves **head, face, neck, arms and legs** exposed. **With the source below the patient, the surgeon's legs receive the most scatter.** Lymphocytes circulating through unshielded areas can show DNA damage after a single endovascular case.

Environmental Factors

- *Distance.* **Inverse square law: doubling the distance cuts dose to one quarter** — one step back from the table matters. All staff should maximise distance.
- *C-arm placement.* Adjust table height so the **detector is close to the patient and the source far** → lower patient dose → less scatter. **Source below the patient** protects the surgeon's chest and abdomen.
- *Instruments.* In **automatic exposure control** mode the machine raises mA and kVp to hold image quality when radiodense objects enter the beam — instruments, cords, and especially the **table's lateral support beam**. Keep them out of the field.

KEY POINTS: FLUOROSCOPY SAFETY

- Patients, surgeons and staff can receive substantial doses from fluoroscopic procedures.
- Use collimation and shielding whenever possible.
- Low-dose and pulsed modes give large reductions with adequate images.
- Track fluoroscopy time and keep dose ALARA.
- Core measures: less time, shielding, more distance.

Concept — *The surgeon's dose is mostly scatter from the patient, so every measure that lowers patient dose — collimation, low-dose and pulsed modes, detector close and source below — also protects the team; time, distance and shielding act on top of that.*

COMPUTED TOMOGRAPHY

Introduction

- Introduced in the early 1970s; **Sir Godfrey Hounsfield and Allan M. Cormack — 1979 Nobel Prize in Physiology or Medicine** (Fig. 5.10). First clinical scan 1971, a brain tumour (Fig. 5.11).
- First scan: **over 5 minutes to acquire, about 2.5 hours to reconstruct**; now abdomen and pelvis in **5–7 seconds** and the heart in a single beat. Principles are unchanged.
- 3D information gives high sensitivity and specificity in benign disease, urological cancer and urogenital trauma.

Computed Tomography Physics

- X-rays are produced **circumferentially**; detectors opposite the source record attenuation at many angles; each angle has a unique profile, and the profiles are **computationally reconstructed** into a 3D volumetric dataset that can be reformatted in any plane.

- **Voxel** = small 3D cube of tissue carrying its attenuation value as a **CT number**. CT numbers are **not absolute** — they vary between machines and within one machine — but remain useful; shown as greyscale and usually viewed as sequential two-dimensional slices.
- Attenuation coefficients are converted to **Hounsfield units (HU)**:

$$HU = [(\mu_{\text{tissue}} - \mu_{\text{water}}) \div \mu_{\text{water}}] \times 1000$$
, where μ = linear attenuation coefficient.
- **Water = 0 HU**; less dense tissue negative, denser positive. **Air: $\mu \approx 0 \rightarrow -1000$ HU**. Range **-1000 to +3000 HU**; low HU displays black, high HU white.
- The eye cannot separate small HU differences across the full range, so a **window** selects an HU range and rescales it, each grey step covering a smaller span.

CT WINDOWS (FIG. 5.12)

Window	Optimised for	Consequence
Soft tissue	Soft-tissue HU	Lung appears completely black
Lung	Low HU	Lung parenchyma discernible
Bone	High HU	Bone and urinary stones ; subtle stones best seen here

Image Quality

- **Image noise**: unwanted variation in pixel values $\rightarrow$ grainy image, poorer contrast resolution. Measured by the **signal-to-noise ratio (SNR)**; **higher SNR = less noise**.
- Noise determinants: **tube current-time product (mAs)** — more dose, less noise; **slice thickness** — thicker slices collect more photons, less noise; **patient size** — larger patients absorb and scatter more, fewer photons reach the detector, more noise; **reconstruction algorithm**.
- **Beam hardening**: low-energy photons are absorbed preferentially, leaving high-energy photons $\rightarrow$ **artificially low HU centrally** or streaks by dense objects. **Metal artefact** from prostheses: dark bands between two metal objects, light or dark streaks from one (Fig. 5.13: bilateral hip implants). **Photon-counting detectors** sort photons into energy bins; high-energy data reduce beam hardening and metal artefact.
- **Partial volume averaging**: a voxel too large contains two tissues and shows their average. Key consequence: **pseudoenhancement of small renal cysts** from adjacent parenchyma, especially on contrast images $\rightarrow$ cyst misread as enhancing tumour $\rightarrow$ overtreatment. Fig. 5.14: cyst 16 HU unenhanced, 39 HU nephrographic — a 23 HU rise, above the 20 HU threshold, yet a benign cyst. Minimise with **narrow collimation, thin slices and a central region of interest**; ultra-high-resolution CT may reduce it.
- **Slice thickness** governs cross-plane resolution and precision of organ measurement; **thinner reconstructed slices increase noise**, which the operator offsets by raising mAs (dose).

Low-Dose and Ultra-Low-Dose CT

- CT is among the most frequent urological investigations and may account for **up to 2% of newly diagnosed cancers in the United States** — hence the drive to cut dose while keeping image quality.

EFFECTIVE DOSES CITED IN THIS CHAPTER

Study	Effective dose
Chest x-ray	0.07 mSv
KUB, single film / two-view	0.7 mSv / 1.4 mSv
Ultra-low-dose CT abdomen/pelvis	1–2 mSv (about 1 mSv with deep learning reconstruction)
Low-dose CT abdomen/pelvis	<3–4 mSv
Conventional CT abdomen/pelvis	10–20 mSv

- Dose is cut by lowering **tube current (mAs)** — **the commonest method** — or tube voltage. Low-dose and ultra-low-dose protocols are routine for **stone diagnosis and follow-up** and spreading to other indications.
- Tube current should be **adjusted to patient size**; tube potential and rotation speed are usually fixed per protocol. **Faster rotation → less blurring; lower tube potential → more contrast but more noise**. Modern scanners **modulate dose in real time** along the absorption profile.

Image Cross-Sections

- **Axial** = horizontal slices; **coronal** = vertical slices separating front from back; **sagittal** = vertical slices separating right from left (Fig. 5.15).
- **Maximum intensity projection** combines slices, placing voxels above an intensity threshold into one image → 3D models; used routinely in **CT angiography** (Fig. 5.16: donor work-up showing three right renal arteries and one left, so the left kidney was taken). Post-processing can strip lower-density parenchyma to display a stone in 3D.

Image Reconstruction

- Ideal reconstruction is **fast, low-dose and high-quality**; most techniques compromise on at least one.
- **SNR** compares signal power with the noise corrupting it; **contrast-to-noise ratio (CNR)** compares mean signal in a tissue of interest with background. Lower SNR and CNR mean poorer images.
- **Filtered back projection (FBP)** — since the 1980s; reconstructs in seconds; **low SNR and CNR, higher dose**.
- **Model-based iterative reconstruction (MBIR)** — computationally heavy; solve for the image → forward-project the attenuation it would produce → compare with measured data → update → repeat until the image fits. **High SNR; about half the dose of FBP with better images** (Fig. 5.17: superior depiction of retroperitoneal fascia and corticomedullary tissue, most obvious at ultra-low dose). **Slow** — limits acute use — and gives a **waxy, plastic appearance**.
- **Hybrid iterative reconstruction (HIR)** — commonly used; vendor-proprietary; statistical modelling with forward projection, **faster than MBIR**, but SNR low like FBP, waxy image, moderate dose.
- **Deep learning reconstruction (DLR)** — AI trained on thousands of matched low-dose/high-dose image pairs through millions of iterations learns to turn noisy images into high-quality ones; **meets all three goals**. Ultra-low-dose DLR images look natural at a **dose close to a plain radiograph**; fast enough for acute settings; supports **metal artefact reduction** software. Needs close work with the manufacturer's application specialists and the local medical physicist. Fig. 5.18: the same patient at 6.3 mSv with HIR and about 1 mSv with DLR plus metal artefact reduction, artefact markedly reduced — below a two-view KUB.

TABLE 5.1 COMPARISON OF CT RECONSTRUCTION TECHNIQUES

Technique	Strengths	Weaknesses	Vendor examples
FBP	Reconstruction speed; control of image characteristics (e.g. kernel); image texture; robust image metrics	Image noise; artefact-prone (streaking); negligible dose reduction ; poor depiction of low-contrast objects	—
MBIR	Noise reduction strong ; low-contrast resolution moderate; dose reduction strong ; artefact reduction strong	Slow; less control of image characteristics; oversmoothing; altered texture; less robust metrics	VEO (GE), FIRST (Canon), IMR (Philips), ADMIRE (Siemens)
HIR	Noise, low-contrast resolution, dose and artefact reduction all moderate	Speed; control of image characteristics; oversmoothing; altered texture; less robust metrics	ASIR/ASIR-V (GE), AIDR 3D (Canon), iDose (Philips), SAFIRE (Siemens)
DLR	Noise reduction, low-contrast resolution, spatial resolution and artefact reduction all strong	Needs large training datasets and accurate selection of training data; slight loss of fine-detail sharpness; less robust metrics	AiCE (Canon), TrueFidelity (GE)

From Agostini A, Borgheresi A, Granata V, et al., Eur Rev Med Pharmacol Sci, 2022.

- Purpose: the trade-off between speed, noise and dose. Pattern: **FBP is fast but noisy and dose-hungry; iterative methods cut noise and dose at the price of speed and a plastic texture; DLR is strongest across the board**. Trap: HIR is faster than MBIR but does not share MBIR's high SNR.

Dual-Energy CT Scan

- **Dual-energy CT (DECT)** acquires two energy levels simultaneously (e.g. **80 and 140 kVp**) and compares attenuation pixel by pixel → material behaviour at each energy, **iodine extraction** (better iodine detection on low-energy images) and **virtual unenhanced images**, which also sidestep motion and breathing artefact between phases.
- Uses: **hyperattenuating cyst versus enhancing renal mass**; adrenal nodules; fewer CT urogram phases (virtual unenhanced replaces the true noncontrast phase); **stone composition — uric acid versus non-uric acid with near-perfect accuracy at low dose**.
- Fig. 5.19: after right radical nephrectomy for T1a clear cell RCC, a small hyperdense left lesion was indeterminate on arterial and nephrographic phases; virtual unenhanced images confirmed hyperdensity, and **40 keV monoenergetic images and an iodine map showed no enhancement → hyperdense cyst**.
- **Limitations**: motion and beam hardening artefact persist; **limited field of view** (obesity); **more radiation than low-dose and ultra-low-dose CT**; longer reconstruction; twice as many images → radiologist workload.

Noncontrast CT

- **Noncontrast CT (NCCT)** uses: diagnosis and follow-up of stones; **first phase of renal-protocol CT and CT urography** (baseline HU); trauma, to delineate extravasation or active haemorrhage on later phases; **renal impairment** — hydronephrosis, lymphadenopathy, larger soft-tissue masses.

Noncontrast CT for Kidney Stones

- Stones attenuate more than parenchyma and urine, so no IV contrast is needed. **NCCT is the gold-standard imaging for urolithiasis** (values in Table 5.2).
- **All stones are visible except protease inhibitor stones (e.g. indinavir)**.
- Gives size, location and **indirect signs of obstruction** (stranding, hydronephrosis), and composition from density: **uric acid 200–400 HU; calcium oxalate 600–1200 HU**.

- **View stones on bone window** — especially residual fragments after PCNL, where a nephrostomy tube or double-J stent of similar density hides them (Fig. 5.21: bone window separates denser stent coils from an encrusting stone).
- Emergency department NCCT use for stones rose several-fold between 1992 and 2009. Against ultrasound, NCCT has higher sensitivity and specificity but costs about twice as much and carries substantial radiation.

GUIDELINES: FIRST IMAGING IN SUSPECTED UROLITHIASIS

Body	Recommendation
AUA Technology Assessment Committee	CT first in adults with symptoms of urolithiasis; standard-dose CT if body mass index (BMI) >30
EAU	Ultrasound first ; CT only if ultrasound inconclusive
AUA surgical guideline	NCCT before PCNL or SWL

- A prospective randomised trial supports the EAU approach: starting with point-of-care or formal ultrasound **reduced radiation with no difference in outcomes**.
- **Limitations:** standard NCCT abdomen/pelvis delivers 10–20 mSv; **10 mSv carries an estimated 1 in 1000 chance of cancer** — a concern in young, recurrent stone formers.

Low-Dose and Ultra-Low-Dose CT for Kidney Stones

- Reducing tube current by **up to 95%** preserves sensitivity, specificity, stone size and HU measurement.

TABLE 5.2 COMPARISONS BETWEEN IMAGING MODALITIES FOR THE EVALUATION OF KIDNEY STONES

Modality	Sensitivity (%)	Specificity (%)	Dose (mSv)	Time (min)	Cost (× KUB)
CT	95	98	10.0	10	10
Low-dose CT	95	97	≈3.0	10	10
Ultra-low-dose CT	94	95	≈1.0	10	10
Ultrasound	84	53	None	20–30	5
KUB	57	76	0.7–1.4	<1	1
Magnetic resonance imaging (MRI)	82	98	None	60	30

Modified from Brisbane W, Bailey MR, Sorensen MD, Nat Rev Urol, 2016.

- Purpose: accuracy against dose, time and cost. Pattern: **the three CT tiers are equally accurate while dose falls tenfold**; ultrasound avoids radiation but has **poor specificity**; KUB is cheapest but **least sensitive**; MRI avoids radiation but is slowest and costliest.
- The table gives conventional CT as 10 mSv; the text gives 10–20 mSv. The table carries a single representative value from a review, the text the usual range.
- Trap: ultra-low-dose CT (about 1 mSv) delivers less than a two-view KUB (1.4 mSv).
- **Obesity:** low-dose protocols struggle; hence standard CT above BMI 30.
- **Body habitus:** ultra-low-dose protocols (**7.5 mAs**) detect ureteric stones well at average weight, but **very large and very thin patients** lose accuracy, which low-dose protocols (**15 and 30 mAs**) preserve. In thin patients the ureter is hard to trace and **phleboliths are confused with ureteric stones**.
- **Also avoid ultra-low-dose CT with metal hardware** (degraded image) and **suspected infected stones** — missing a 1–2 mm ureteric stone is riskier than the radiation of a conventional scan. DLR may rescue ultra-low-dose quality in these groups.

CT With Intravenous Contrast

- Contrast-enhanced abdominal CT is the **commonest modality for detecting and characterising cystic and solid renal tumours**; contrast improves depiction of organs, lesions and soft tissue.
- **Phase** = time point after injection; **protocol** = the set and order of phases for a given question. Protocols vary by institution but share the basic phases.

CONTRAST PHASES

Phase	Timing	What is enhanced	Use
Early arterial	20–30 s	Systemic arteries after pulmonary transit	Renal arteries, variants, active arterial bleeding, stenosis, aneurysm
Corticomedullary (late arterial)	30–70 s	Cortex, not medulla; arteries and veins	Cortical and hypervascular masses; renal arterial anatomy
Hepatic portal (portal venous)	70–90 s	Portal vein, liver parenchyma, veins	Routine abdomen CT; liver, lymph nodes, soft tissue
Nephrographic	90–120 s	Tubules and medulla; corticomedullary differentiation lost	Renal masses — some small medullary masses seen only here
Delayed (excretory)	7–10 min	Collecting system and ureters	Filling defects, masses, extravasation

- **Enhancement** = increase in density with contrast; its extent and timing distinguish pathology, e.g. RCC subtype. **Washout** = decrease against a prior phase, reflecting how fast contrast clears.

Routine Abdomen and Pelvis CT With Intravenous Contrast

- Emergency acute abdomen, ± chest CT; **always includes the hepatic portal phase** (soft tissue, bowel, lymphadenopathy). Staging and surveillance of urological cancers, including testicular; combined with other phases for bladder and kidney cancer; postoperative bowel obstruction, urine leak and collections; ± oral or rectal contrast.
- **High-risk prostate cancer nodal staging**: positivity thresholds range 0.5–2 cm, but **10 mm is the widely accepted cut-off**; sensitivity is poor, so **prostate-specific membrane antigen (PSMA) positron emission tomography (PET)-CT** is displacing it.
- **Testicular germ cell tumour after orchidectomy**: abdomen–pelvis CT for retroperitoneal nodes; surveillance means repeated scans and cumulative dose; **low-dose contrast CT is as effective as standard dose in stage 1** (prospective study).

CT Angiogram

- Uses the early arterial phase: renal artery aneurysm, stenosis, occlusion; **living kidney donor work-up for accessory or aberrant vessels**, often as a **four-phase CT (noncontrast, arterial, nephrographic, excretory)**; acute mesenteric ischaemia; **active arterial bleeding** as extravasation in haematoma or trauma.
- **Suspected renal trauma protocol**: noncontrast + arterial + **nephrographic** (parenchyma) + **delayed** (collecting system integrity).

Adrenal Mass Protocol

- Three phases: **noncontrast, portal venous, and delayed at 15 minutes** for washout.
- **Lipid-rich adenoma**: **<10 HU unenhanced → benign**. Lipid-poor adenomas need washout calculation; **adenomas wash out faster** than other adrenal lesions.

ADRENAL WASHOUT

	Formula	Favours adenoma
Absolute	$(\text{HU portal venous} - \text{HU delayed 15 min}) \div (\text{HU portal venous} - \text{HU unenhanced}) \times 100$	>60%
Relative	$(\text{HU portal venous} - \text{HU delayed 15 min}) \div \text{HU portal venous} \times 100$	>40%

- The two share a numerator; the **denominator differs** — **absolute washout needs the unenhanced value**.

Renal Mass Protocol

- More CT for any abdominal symptom → more **incidental RCC**; an incidental renal mass is found on a substantial minority of abdominal CT scans.
- Phases: **noncontrast, corticomedullary, nephrographic**, ± delayed (Fig. 5.22: a small RCC barely visible unenhanced, vague in corticomedullary, clear in nephrographic and delayed). The noncontrast phase gives baseline density.

RENAL MASS ENHANCEMENT THRESHOLDS

Change in attenuation after contrast	Interpretation
≥20 HU	Enhancing — vascularised solid tumour
10–20 HU	Equivocal
<10 HU	Non-enhancing — cyst

- Trap: **cystic lesions under 10–15 mm may falsely enhance** through volume averaging.
- **Nephrographic phase is the most sensitive for detecting renal tumours**, because nearly all masses enhance less than normal parenchyma; it also confirms enhancement and shows **tumour thrombus extent into the inferior vena cava**.
- **Corticomedullary phase**: cortical masses, **clear cell RCC (ccRCC) versus other subtypes**, arterial anatomy, renal vein involvement.
- **Subtype**: ccRCC enhances **early and avidly**; papillary, chromophobe and oncocytoma enhance **later**. BMI and cardiac output distort enhancement; an **aorta-corrected attenuation** (tumour HU × ratio of cohort mean aortic HU to the patient's aortic HU) improves discrimination in the corticomedullary phase.
- **Angiomyolipoma (AML)**: **macroscopic fat ≤ -10 HU on NCCT → diagnostic** (Fig. 5.24: 8 cm left upper-pole AML with fat matching retroperitoneal and subcutaneous fat, plus vascular and smooth-muscle components). Fat-poor AML: **hyperdense on NCCT with rapid enhancement and washout** supports the diagnosis.
- Cystic masses are classified by the **2019 Bosniak classification** (Table 5.3, Fig. 5.23).

TABLE 5.3 2019 BOSNIAK CLASSIFICATION OF RENAL CYSTIC TUMOURS

Class	CT findings	Malignancy / management
I	Well defined, homogeneous simple fluid -9 to 20 HU ; smooth thin wall (≤2 mm) , may enhance; no septa, no calcification	Benign simple cyst — no follow-up
II	(a) Cystic mass, smooth thin wall, few (1–3) thin (≤2 mm) septa , wall/septa may enhance, calcification of any type; <i>or homogeneous masses</i> : (b) ≥70 HU on noncontrast CT ; (c) non-enhancing >20 HU at renal-mass protocol, calcification of any type; (d) -9 to 20 HU on noncontrast CT ; (e) 21–30 HU on portal-phase contrast CT ; (f) hypodense, too small to characterise. A cystic mass has <about 25% enhancing tissue on visual inspection	Benign or highly likely benign — no follow-up

Class	CT findings	Malignancy / management
IIF	Smooth minimally thickened (3 mm) enhancing wall or septa, or many (≥4) smooth thin enhancing septa	Likely benign, malignancy <10% — follow-up at 6 and 12 months, then annually for 5 years
III	≥1 thick (≥4 mm) enhancing wall/septum, or irregular wall/septa: ≤3 mm convex protrusion with obtuse margins	Intermediate, about 50% — refer to urologist
IV	≥1 enhancing nodule : convex protrusion of any size with acute margins , or ≥4 mm with obtuse margins	Often malignant, about 90% — refer to urologist

Measurements on contrast-enhanced CT, perpendicular to the wall or septum, excluding it. From Tsili AC, Varkarakis I, Pasoglou V, et al., Eur J Radiol, 2023.

- Purpose: triage of cystic masses into ignore, follow, or refer. Pattern: **wall and septal thickness climbs 2 → 3 → ≥4 mm**; septal **number** climbs 1–3 → ≥4; the protrusion rule separates III from IV.
- Trap: a **≤3 mm obtuse protrusion is irregularity (III)**; **≥4 mm obtuse or any acute-angled protrusion is a nodule (IV)**. Calcification of any type is compatible with class II.

CT Urography

- **CT urogram (CTU)** images kidneys, ureters and bladder with thin slices and **always includes a delayed excretory phase**.

CT UROGRAPHY TECHNIQUES

Technique	Acquisition	Trade-off
Single-bolus (standard)	Noncontrast (stones, baseline HU) → corticomedullary → nephrographic → excretory	Most phases, most radiation
Split-bolus	Two injections → one combined nephrographic-excretory phase	Less radiation, but lower sensitivity for subtle urothelial lesions and small RCC , possible understaging; may need more contrast → acute tubular necrosis risk
DECT	Virtual unenhanced phase from contrast phases; with split-bolus → all phases in one acquisition (Fig. 5.25)	Least radiation for full information

- **Upper tract urothelial carcinoma (UTUC): CTU is the routine diagnostic test**, with high sensitivity and specificity in meta-analysis. Appearance: soft-tissue density on noncontrast images, **lower attenuation than stone or clot**; early moderate enhancement; **filling defect in the excretory phase** (Fig. 5.26).
- Distal ureters often fail to opacify even when normal. Aids: IV diuretic, oral or IV hydration, prone position, abdominal compression — **hydration is the simplest and most effective**.
- **Bladder cancer:** CTU recommended for **multiple or high-risk tumours or trigonal involvement**. Tumour enhances more than normal wall: polypoid mass, broad-based plaque-like thickening, or diffuse infiltration (Fig. 5.27). High sensitivity and specificity for primary and recurrent tumour, provided the **bladder is well distended** and the wall is read on axial plus coronal and sagittal reformats.
- **Local staging:** macroscopic perivesical fat invasion (**T3b**) and adjacent organs (**T4**); accuracy rises with stage. Perivesical invasion: **irregular tumour-fat interface**, often with stranding. **Perform CT before transurethral resection of bladder tumour (TURBT)**, because post-resection change mimics invasion.
- **CT cannot reliably separate muscle-invasive from non-muscle-invasive disease**; bladder wall retraction and hydronephrosis are indirect signs of invasion.

- **Nodes in bladder cancer:** moderate accuracy; pathological if **short axis >8 mm pelvic, >10 mm abdominal**.
- **Ureteric injury** (traumatic or iatrogenic): delayed-phase extravasation locates it. **Laceration → extravasation with distal ureter opacified; transection → distal ureter not opacified** (Fig. 5.28: mid-ureteric leak after colostomy takedown). Excretion begins **3–4 minutes** after injection and lasts over an hour; with obstruction (e.g. pelviureteric junction obstruction) it is delayed, so **add later excretory images**.

CT Cystography

- **Modality of choice for traumatic, iatrogenic and other bladder pathology;** trauma patients already having abdomen–pelvis CT get it in the same sitting; used less often in isolation for fistulas and leaks.
- **Technique:** exclude urethral injury → **at least 300–350 mL dilute contrast via catheter**. **Passive filling by excreted contrast is inadequate;** retrograde filling is required (Fig. 5.29: passive filling missed an extraperitoneal rupture that retrograde filling showed). **No postdrainage images needed**, unlike conventional cystography.
- Done properly, conventional and CT cystography are both highly accurate.

EXTRAPERITONEAL VERSUS INTRAPERITONEAL BLADDER RUPTURE

Extraperitoneal	Intraperitoneal
Flame-shaped or molar tooth collection around the bladder in the space of Retzius ; if the urogenital diaphragm is disrupted, contrast extends into the scrotum	Contrast in the peritoneal cavity outlining bowel loops and surrounding liver and/or spleen (Fig. 5.30: dome rupture)

- **Pitfalls:** a large pelvic haematoma can hide extravasation from a dome rupture; **diverticula mimic extravasation**; a **large extraperitoneal rupture prevents distension and so hides a coexisting intraperitoneal rupture**.

Positron Emission Tomography CT

- **PET-CT** images biochemical processes (e.g. glucose metabolism). Radioisotopes emitting **positrons** are coupled to ligands (e.g. glucose) taken up avidly by tumour → positron meets electron → **annihilation → two 511 keV photons in opposite directions** → detected → 3D map of isotope distribution overlaid on CT (Fig. 5.31).
- **PSMA PET-CT** in prostate cancer: tracers **gallium-68 (^{68}Ga)–PSMA** and **fluorine-18 (^{18}F)–PSMA**; uptake quantified as the **standardised uptake value (SUV)**, which accounts for injected activity, body weight and injection-to-imaging time.
- Better detection of locoregional and metastatic disease (Fig. 5.32: thoracic and abdominal nodes). **Biochemical recurrence (BCR)** detected earlier: **positivity rises with prostate-specific antigen (PSA) level**, detection is useful even at **PSA <0.5 ng/mL**, with very high specificity; **^{18}F -PSMA outperforms ^{68}Ga -PSMA for BCR**.
- Metastatic detection: **PSMA PET-CT is more sensitive than MRI, choline PET and bone scan**, and has the highest sensitivity and specificity for skeletal metastases.
- **RCC:** carbonic anhydrase IX is expressed on almost all ccRCC; **zirconium-89 (^{89}Zr)–girentuximab PET-CT** (antibody to carbonic anhydrase IX) diagnoses localised ccRCC and may separate ccRCC from non-ccRCC in metastatic disease.

KEY POINTS: COMPUTED TOMOGRAPHY

- A rotating x-ray tube acquires data that are reconstructed into slices of 3D anatomy.

- NCCT is highly sensitive for urinary stones and is the first phase of several protocols.
- Low-dose and ultra-low-dose CT diagnose stones effectively and reduce dose by up to 95%.
- Contrast improves tissue differentiation; timing of enhancement and washout identifies pathology.
- CTU uses IV contrast and thin slices in the excretory phase to image the urinary tract.
- PET-CT uses labelled molecules taken up by metabolically active tissue to highlight possible neoplasia.

Concept — CT answers two questions in sequence: **baseline HU says what a lesion is made of (fat, fluid, stone, blood), and the change in HU across timed phases says what it does (enhances, washes out, excretes, leaks).** Each protocol is simply the set of phases that answers a given question at the lowest dose.

CONTRAST MEDIA

- Contrast improves differentiation of adjacent structures; agents are safe for most patients, but **clinicians and staff must recognise and treat reactions, with drugs and equipment at hand.**
- **Severe allergic reactions to intravascular iodinated contrast are rare.**

HYPERSENSITIVITY REACTIONS TO IODINATED CONTRAST

Immediate	Nonimmediate (delayed)
Within the first hours : erythema, urticaria, angio-oedema, hypotension, dyspnoea, nausea, vomiting; anaphylaxis if severe	Hours to days , generally milder: maculopapular rash, urticaria

- **Observe for at least 30 minutes** after iodinated contrast.
- **Management:** stop contrast → anaphylaxis: **intramuscular (IM) adrenaline (epinephrine) 0.3–0.5 mg every 5–15 minutes, to a maximum of 1 mg** → oxygen and β_2 -agonist (salbutamol/albuterol) **2 puffs of 90 µg, repeatable up to 3 times** → mild reactions (pruritus): observation or H_1 -antagonist, with close monitoring because they can progress → later, **drug provocation and skin tests** to confirm contrast as the cause → careful risk–benefit analysis before any further iodinated contrast.
- **Premedicate high-risk patients:** previous contrast reaction, asthma, atopy, severe cardiovascular disease, drug allergy, female sex. Breakthrough reactions still occur, and benefit against rare severe reactions is debated, but premedication remains recommended.

BOX 5.2 PREMEDICATION IN PREPARATION FOR CONTRAST ADMINISTRATION

Regimen	Medication	Dose	Timing before contrast
1	Prednisone	50 mg orally (PO)	13 h, 7 h and 1 h
	Diphenhydramine	50 mg PO or IM	1 h
2	Methylprednisolone	32 mg PO	12 h and 2 h
	Diphenhydramine	50 mg PO or IM	1 h

Modified from ACR Committee on Drugs and Contrast Media, 2023.

- Purpose: two interchangeable steroid regimens, each with an antihistamine 1 hour before. Trap: prednisone is **three doses (13–7–1)**, methylprednisolone **two (12–2)**.

CONTRAST-ASSOCIATED VERSUS CONTRAST-INDUCED ACUTE KIDNEY INJURY

Contrast-associated AKI (CA-AKI)	Contrast-induced AKI (CI-AKI)
Formerly <i>postcontrast</i> AKI	Formerly <i>contrast-induced nephropathy</i>
AKI within 48 hours of contrast, whether or not contrast caused it	Subset in which contrast is the causative agent

- **Kidney Disease: Improving Global Outcomes (KDIGO) definition of acute kidney injury (AKI)**, any one of: serum creatinine rise ≥ 0.3 mg/dL (≥ 26.5 μ mol/L) **within 48 hours**; rise $\geq 50\%$ ($\geq 1.5 \times$ baseline) **within 7 days**; urine output < 0.5 mL/kg/hour for ≥ 6 hours.
- Some evidence suggests modern agents do not significantly raise AKI risk; the **American College of Radiology (ACR) Committee on Drugs and Contrast Media** holds that CI-AKI is **rare but real**.
- **Prevention**: noncontrast study where possible; **IV fluids for high-risk patients**; iso-osmolar or low-osmolar contrast; **kVp lowered to 80** so iodine attenuates optimally and less contrast is needed; weight-based dosing.

KEY POINTS: CONTRAST MEDIA

- Contrast aids tissue differentiation but carries rare adverse reactions.
- Give adrenaline for severe immediate reactions; add oxygen and β_2 -agonists for bronchospasm.
- Observe mild reactions or give H_1 -antagonists, with close monitoring.
- Premedicate patients at high risk of allergic reaction.
- For patients at risk of CI-AKI, give IV fluids and reduce contrast dose.

Concept — Contrast carries **two unrelated risks**: hypersensitivity, which is idiosyncratic and handled by premedication and adrenaline, and nephrotoxicity, which is dose-related and handled by hydration and less contrast. Prophylaxis for one does nothing for the other.

ARTIFICIAL INTELLIGENCE

- Models used: machine learning, artificial neural networks and **convolutional neural networks (CNNs)** — **the commonest in diagnostic imaging**. CNNs detect stones on CT with very high sensitivity; artificial neural networks detect renal tumours with good accuracy.
- Main advantage: **processing big data rapidly**. Growing electronic records supply training data; pattern recognition improves the systems.
- AI may speed interpretation and improve accuracy, but **replacing radiologists is distant**; its role in imaging will grow.